

Interplay between Evolutionary and Epidemic Timescales Challenges the Outcome of Control Policies

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The classical SIR model, assuming constant viral traits, represents the cornerstone model for computing key indicators during epidemic outbreaks, such as the expected peak of infections or the impact of control policies. Viral evolution has been reported to challenge the physics of the SIR model, changing the nature of the epidemic transitions or the early-time dynamics of outbreaks. Here we consider a minimal extension of the SIR model, allowing infectiousness to evolve, to explore how the latter mechanism affects the two aforementioned indicators. We show that evolution induces a nonmonotonic behavior of the epidemic peak with the basic reproduction number and undermines the impact of control policies, as lifting interventions too early can lead to worse epidemic scenarios than no action. We derive analytical expressions for the critical mutation rate and intervention time governing this behavior and identify a strong asymmetry between control strategies: while shortening the infectious period hinders transmission without suppressing the evolution of viral infectiousness, lowering transmission both reduces cases and slows down this evolution.

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Introduction—The dynamics of infectious diseases emerge from the interplay between pathogen traits and host populations [1,2]. Classical epidemic models, such as the susceptible-infected-recovered (SIR) framework [3,4], have been instrumental in understanding outbreak dynamics and in guiding the design of control strategies during recent pandemics [5–8]. Recent theoretical studies [9–14], and particularly the work of Morris *et al.* [15], have further analyzed how simple, fixed-strength interventions can be robust and near-optimal control strategies.

A key underlying assumption in these approaches is that pathogen traits remain effectively constant over the course of an outbreak. Yet, many pathogens evolve on timescales comparable to epidemic spreading, so that mutation can directly affect epidemic dynamics [16–20]. The impact of control policies on viral evolution has been mostly addressed in two-strain settings, characterizing analytically how interventions alter the emergence, survival and immune escape probabilities of mutants both within hosts [21–24] and at the population level [25–27]. Beyond the two-strain settings, phylodynamic approaches have proven substantial alterations in the evolutionary trajectories of Ebola and Influenza A viruses following nonpharmaceutical

interventions applied to contain Ebola [28] or COVID-19 outbreaks, respectively [29,30].

The theoretical study of how viral evolution shapes epidemic trajectories in multistrain settings has been much less explored in the literature [31,32]. It has been reported that evolution induces superexponential growth of outbreaks or abrupt epidemic transitions [31] in the SIR model; nonetheless, the impact of evolution on pivotal indicators such as the peak of infections or the impact of control policies remains a theoretical challenge. In this Letter, we address these questions within a minimal extension of the SIR model in which infectivity is an evolving pathogen trait [31]. We show that evolution reverses the monotonic dependence of the epidemic peak on the basic reproduction number and the short-term benefits of epidemic control policies. We also find strongly asymmetric outcomes of interventions shaping within-host or inter-host dynamics under evolution. Our results therefore show that epidemic control and pathogen evolution are intrinsically coupled and should therefore be studied jointly for rapidly evolving viruses.

Modeling epidemics with infectivity evolution—We consider a minimal extension of the SIR model in which individuals are either susceptible (S), infected (I), or recovered (R), and where infectivity is an evolving trait [see Fig. 1(a)]. For contagions, we assume that infected individuals make k contacts per time step, transmitting their

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associated strain to a susceptible contact with a probability λ , hereafter referred to as infectivity of that strain. We assume a competitive exclusion regime, therefore neglecting coinfections by multiple strains. Following [31–34], infectivity evolves in a one-dimensional trait space via symmetric mutations of size $\pm\Delta\lambda$ occurring at rate D . Finally, infected individuals recover at a rate μ , entering into the R compartment. The epidemic state of our system is characterized by the probability density of infected individuals across the trait space $\rho_I(\lambda, t)$ and the global fraction of recovered $r(t)$ individuals. Taking into account the rules described above, the time evolution of these quantities read,

$$\frac{\partial \rho_I(\lambda, t)}{\partial t} = k\lambda \rho_I(\lambda, t)s(t) - \mu \rho_I(\lambda, t) + \frac{D}{2} \frac{\partial^2 \rho_I(\lambda, t)}{\partial \lambda^2}, \quad (1)$$

$$\frac{dr(t)}{dt} = \mu i(t), \quad (2)$$

where $i(t) = \int d\lambda \rho_I(\lambda, t)$ and $s(t) = 1 - i(t) - r(t)$ are the disease prevalence and the fraction of susceptible individuals at time t , respectively. For mathematical convenience, we have used an effective diffusion rate $\mathcal{D} = D(\Delta\lambda)^2$, which remains invariant under rescalings of D and $\Delta\lambda$ that keep $D(\Delta\lambda)^2$ constant (see Supplemental Material Note 1 [35]).

Equation (1) is a reaction-diffusion equation in trait space, structurally equivalent to replicator-mutator dynamics under time-dependent selection coefficients (see Supplemental Material Note 1 [35]). In particular, the first term selects for strains with larger λ through transmission, mediated by the depletion of the pool of susceptible individuals $s(t)$ throughout the dynamics. Conversely, the last term shows how mutation dynamics is translated into a diffusion of infected individuals in the trait space.

Effect of infectivity evolution on epidemic trajectories—To characterize the epidemic trajectories, we monitor the time evolution of both the epidemic prevalence $i(t)$ and the associated basic reproduction number defined as

$$\mathcal{R}_0(t) = \frac{k\bar{\lambda}(t)}{\mu}, \quad (3)$$

with the average infectivity

$$\bar{\lambda}(t) = \frac{\int d\lambda \lambda \rho_I(\lambda, t)}{i(t)}. \quad (4)$$

Unless otherwise stated, both quantities are obtained from the numerical integration of Eqs. (1) and (2), assuming an initial condition where a single strain with basic reproduction number $\mathcal{R}_0 = 2$ affects a tiny fraction $i_0 = 10^{-3}$ of the population, with $\mu = 1/7$ days $^{-1}$ and $k = 10$ contacts per day. For evolution, we assume a discrete representation of the trait space with $\Delta\lambda = 10^{-3}$ (see Supplemental Material Note 1 [35]).

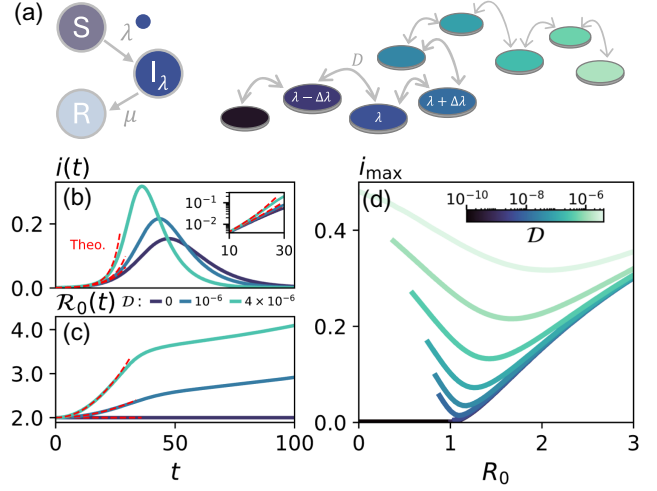


FIG. 1. Effect of infectivity evolution on epidemic trajectories. (a) Schematic illustration of the epidemic model with evolving infectivity. (b),(c) Temporal evolution of the epidemic prevalence $i(t)$ (b) and basic reproduction number $\mathcal{R}_0(t)$ (c). (d) Epidemic peak $i_{\max}$ as a function of the initial basic reproduction number $\mathcal{R}_0$ for different values of $\mathcal{D}$, revealing an abrupt epidemic transition induced by infectivity evolution. In panels (b) and (c), dashed lines show the theoretical predictions from Eqs. (6) and (7), and we set $\mathcal{R}_0 = 2$. In panels (b), (c), and (d), we set $i_0 = 10^{-3}$, $\mu = 1/7$, $k = 10$ and $\Delta\lambda = 10^{-3}$, and stop simulations if $i(t) < 10^{-4}$.

Infectivity evolution qualitatively reshapes epidemic trajectories, inducing a superexponential growth, advancing the epidemic peak and increasing its magnitude [Fig. 1(b)]. Moreover, Fig. 1(c) shows that infectivity evolution presents an initial superlinear growth followed by a bending which is slowed down at later stages. Both results can be understood analytically from the dynamics of the average infectivity. In Supplemental Material Note 2 [35], we show that differentiating Eq. (4) leads to

$$\frac{d\bar{\lambda}}{dt} = k s(t) \text{Var}(\lambda), \quad (5)$$

where $\text{Var}(\lambda)$ is the variance of λ values considering the prevalence distribution across traits $\rho_I(\lambda, t)$. The diffusive contribution yields $\text{Var}(\lambda) \simeq \mathcal{D}t$. For early-time dynamics, one can assume that $s(t) \simeq 1$. Combining both assumptions, we can integrate Eq. (5), yielding

$$\bar{\lambda}(t)|_{t \rightarrow 0} = \lambda_0 + \frac{1}{2} k \mathcal{D} t^2, \quad (6)$$

which directly translates into a superexponential growth of the prevalence,

$$i(t)|_{t \rightarrow 0} = i_0 \exp \left[\mu(\mathcal{R}_0 - 1)t + \frac{k^2 \mathcal{D}}{6} t^3 \right]. \quad (7)$$

Note that infectivity evolution introduces an additional cubic term [31] in the growth exponent compared to the

standard SIR model. Equations (6) and (7) accurately capture the early-time dynamics in Figs. 1(b) and 1(c). As the epidemic progresses, the susceptible depletion slows down the growth of $\bar{\lambda}(t)$, producing the bending observed in Fig. 1(c) (see Appendix A for further details).

Effect of infectivity evolution on epidemic control—The epidemic peak $i_{\max}$ is a pivotal quantity for mathematical epidemiology, as it indicates the highest pressure on health systems during epidemic outbreaks. Figure 1(d) shows how the epidemic transition of the SIR model is strongly altered by evolution. While in the absence of evolution, i.e., $D = 0$, epidemic outbreaks occur through a second-order transition at $\mathcal{R}_0 = 1$, increasing D gives rise to abrupt epidemic transitions occurring for lower $\mathcal{R}_0$ values. This occurs because infectivity evolution enhances transmissibility during the outbreak, effectively pushing the dynamics into the supercritical region of the classical SIR phase diagram even when starting from marginally transmissible strains [31]. Note that the transition at lower finite $\mathcal{R}_0$ values appears because we assume finite populations and set a prevalence threshold $i(t) = 10^{-4}$ below which the outbreak becomes extinct; in infinite populations, the active phase extends down to $\mathcal{R}_0 = 0$ for finite D (see Supplemental Material Note 3 [35]).

Moreover, Fig. 1(d) also reveals the nonmonotonic dependence of $i_{\max}$ with $\mathcal{R}_0$, meaning that mildly infectious pathogens can lead to worse epidemic scenarios due to evolution. This occurs because those viruses start evolving at low prevalence values for longer times without depleting the pool of susceptible individuals, allowing the infectivity distribution to broaden and generate high- λ variants that are later selected by transmission (see Supplemental Material Note 3 [35]).

The latter result represents a cautionary tale for the short-term benefits of control policies, as lowering the basic reproduction number can lead to a worse epidemic scenario. We now tackle this issue by addressing the impact of control policies on the epidemic curves of our model. Our interventions are characterized by two parameters [15]: the duration of the intervention τ and its strength ε . Namely, our intervention rescales the reproduction number, i.e., $\mathcal{R}_0(t) \rightarrow \varepsilon \mathcal{R}_0(t)$, with $\varepsilon \in (0, 1)$, over the intervention time τ . Equation (3) shows that such goal can be achieved through three different strategies: (i) λ control (reducing transmissibility as $\lambda \rightarrow \varepsilon \lambda$), (ii) k control (reducing contacts as $k \rightarrow \varepsilon k$), and (iii) μ control (shortening the infectious period via $(\mu \rightarrow \mu/\varepsilon)$). While the first two strategies are related to interhost transmission events, the last intervention aims at accelerating viral clearance within hosts. Unless otherwise stated, we set $\varepsilon = 0.6$ and treat k control and λ control jointly, which are equivalent within the model (see Supplemental Material Note 4 [35]).

In the absence of infectivity evolution, all policies monotonically reduce the epidemic peak as the intervention duration τ increases [see Fig. 2(a) for k or λ control and

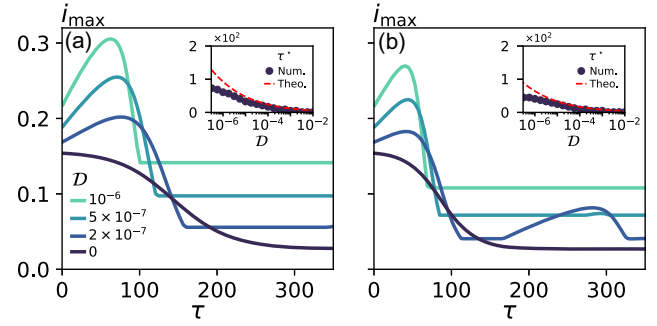


FIG. 2. Effect of infectivity evolution on epidemic control. (a) Epidemic peak $i_{\max}$ under k or λ control as a function of the intervention time τ for different values of the effective diffusion rate D . (b) Same as in (a) for μ control. In both panels, inset represents the theoretical (red dashed line) prediction obtained with Eq. (11) and the numerical (blue dots) values for the peak of the critical intervention time yielding the highest $i_{\max}$ value. In both panels we set $i_0 = 10^{-3}$, $\mathcal{R}_0 = 2$, $\mu = 1/7$, $k = 10$, and $\varepsilon = 0.6$.

Fig. 2(b) for μ control], implying that longer interventions are always beneficial. When evolution is considered, however, $i_{\max}$ becomes a nonmonotonic function of τ , initially increasing and then decreasing after reaching a maximum. This nonmonotonic dependence for short durations of the control policies also appears for the attack rate r_{∞} (see Supplemental Material Note 5 [35]). Consequently, lifting a control policy too early may result in a larger epidemic peak than in the absence of intervention ($\tau = 0$). Note that all the curves eventually converge to plateaus, indicating that the peak of contagions $i_{\max}$ occurs during the controlled phase and not after lifting the intervention.

The nonmonotonic dependency of $i_{\max}$ on τ can be captured analytically. As derived in Supplementary Material Note 4 [35], the early-time evolution of the average infectivity under intervention is

$$\bar{\lambda}(t, \varepsilon)|_{t \rightarrow 0} = \lambda_0 + \frac{1}{2} \alpha k D t^2, \quad (8)$$

where $\alpha = \varepsilon$ for k, λ control, and $\alpha = 1$ for μ control. As a consequence, the prevalence dynamics follows

$$i(t, \varepsilon)|_{t \rightarrow 0} = i_0 \exp \left[\frac{\mu \alpha}{\varepsilon} (\varepsilon \mathcal{R}_0 - 1) t + \frac{\alpha^2 k^2}{6} D t^3 \right]. \quad (9)$$

We can estimate the peak observed after lifting the intervention at $t = \tau$, hereafter denoted by $i_{\max}(\tau)$, by introducing $i(\tau)$ and $\lambda(\tau)$ in the usual expression of the epidemic peak of the SIR model (see Appendix B and Supplemental Material Note 4 [35] for details). Differentiating $i_{\max}$ with respect to τ yields:

$$\frac{di_{\max}}{d\tau} = -\mu i(\tau) \alpha \frac{(1 - \varepsilon)}{\varepsilon} + \frac{\alpha k^2}{\mu} D \tau \frac{\ln(\mathcal{R}_0(\tau) s(\tau))}{\mathcal{R}_0(\tau)^2}. \quad (10)$$

This expression makes explicit the competition between epidemiological suppression (first term) and evolutionary amplification (second term) for short intervention durations. In particular, given an intervention duration τ , the sign of the derivative changes at a critical diffusion strength $\mathcal{D}_c$ (detailed explicitly in Appendix C and in Supplemental Material Note 4 [35]), what pinpoints that interventions improve (worsen) the epidemic scenario for viruses with $\mathcal{D} < \mathcal{D}_c$ ($\mathcal{D} > \mathcal{D}_c$).

Considering $\mathcal{D} > \mathcal{D}_c$, we can also estimate the duration of the intervention τ^* producing the worst epidemic scenario, i.e., the maximum value of the epidemic peak $i_{\max}$. Imposing $(di_{\max}/d\tau)|_{\tau=\tau^*} = 0$ yields (see Supplemental Material Note 4 [35] for details)

$$\tau^* \simeq \left[-\frac{1}{3B\mathcal{D}} W_{-1} \left(-\frac{3B\mu^3(1-\varepsilon)^3 i_0^3}{\varepsilon^3 C^3 \mathcal{D}^2} \right) \right]^{1/3}, \quad (11)$$

with $B = \alpha^2 k^2 / 6$ and $C = k^2 \ln \mathcal{R}_0 / (\mu \mathcal{R}_0^2)$. W_{-1} denotes one of the branches of the Lambert function W , which fulfills $W(z)e^{W(z)} = z$. The resulting analytical predictions are in quantitative agreement with numerical simulations (see insets in Fig. 2), particularly for large values of $\mathcal{D}$, where the characteristic dynamical timescale is reduced due to the evolution of the virus under control and, therefore, evolution after lifting the intervention can be neglected. The latter does not occur for lower $\mathcal{D}$ values, where postlifting evolutionary changes crucially determine τ^* values. For completeness, in Supplemental Material Note 6 [35] we assess the influence of the basic reproduction number $\mathcal{R}_0$, the policy strength ε , the intervention activation threshold (both triggered by time delay or prevalence), and a possible negative bias toward lower infectivity values, showing that the nonmonotonic behavior is robust.

Asymmetric effect of control policies shaping interhost or within-host dynamics—Comparing Figs. 2(a) and 2(b), we observe a difference between k, λ control, and μ control. In the absence of infectivity evolution, i.e., the usual SIR model, shortening the infectious period yields a smaller postlifting peak than reducing transmissibility. We observe that infectivity evolution amplifies this effect for short interventions, whereas it might revert it for longer interventions, as highlighted by the secondary peak appearing for $\mathcal{D} = 2 \times 10^{-7}$ in Fig. 2(b).

The impact of the difference between interventions is made explicit in Fig. 3(a), which shows $\Delta i_{\max} = i_{\max}^{(k,\lambda)} - i_{\max}^{(\mu)}$. Considering a constant $\mathcal{D}$ value, e.g., $\mathcal{D} = 10^{-6}$, we first observe that increasing τ yields a transition from a similar outcome of both interventions ($\Delta i_{\max} \simeq 0$) to a substantially larger epidemic peak after k, λ control compared to the one observed after μ control. Note that this transition also appears in absence of evolution, i.e., very low $\mathcal{D}$ values, but evolution exacerbates it and anticipates its onset.

To gain insights into this transition, we represent the epidemic trajectories $i(t)$ along with the evolution of the

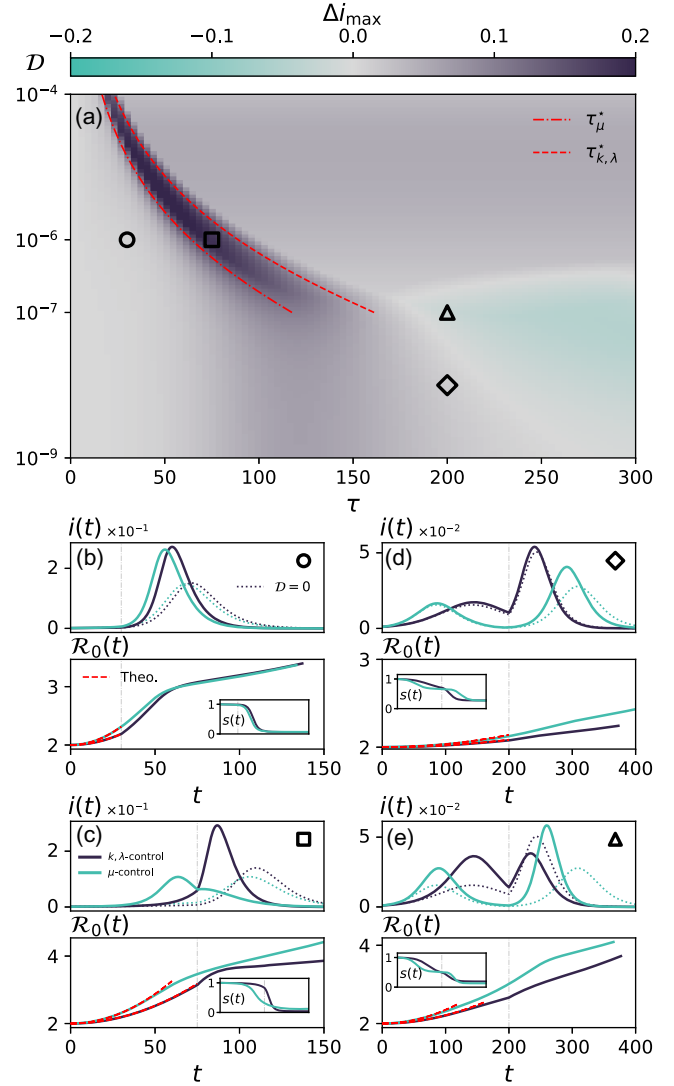


FIG. 3. Asymmetric effect of control policies. (a) Difference in epidemic peak $\Delta i_{\max}$ between the k, λ control and μ control as a function of τ and $\mathcal{D}$. Dashed curves indicate the theoretical predictions for the critical intervention durations τ^* associated with each strategy. (b)–(e) Temporal evolution of the epidemic prevalence $i(t)$ (top row) and the corresponding time-varying basic reproduction number $\mathcal{R}_0(t)$ (bottom row) for (b) $\tau = 30$ and $\mathcal{D} = 10^{-6}$, (c) $\tau = 75$ and $\mathcal{D} = 10^{-6}$, (d) $\tau = 200$ and $\mathcal{D} = 10^{-8}$, and (e) $\tau = 200$ and $\mathcal{D} = 10^{-7}$. Panels (b)–(e) include the theoretical prediction for the early-time dynamics (red dashed line) and, in the inset, the evolution of the susceptible population $s(t)$. Moreover, we stop simulations if $i(t) < 10^{-4}$. In all panels, we set $i_0 = 10^{-3}$, $\mathcal{R}_0 = 2$, $\mu = 1/7$, $k = 10$, and $\varepsilon = 0.6$. Colors distinguish control strategies, as indicated in panel (c), and vertical lines indicate lifting times. Symbols in panel (a) correspond to the parameter choices shown in panels (b)–(e).

basic reproduction number $\mathcal{R}_0(t)$ and the pool of susceptible individuals $s(t)$ for $\tau = 30$ days [Fig. 3(b)] and $\tau = 75$ days [Fig. 3(c)]. Comparing the curves of susceptible individuals, we can observe how the acceleration of the epidemic time scales triggered by μ control leads to a faster

depletion of the pool of susceptible individuals. While this faster depletion is not substantial for very short interventions [Fig. 3(b)], it becomes evident once the peak of the μ control epidemic curves has been reached [Fig. 3(c)]. In that case, the exhausted pool of susceptible individuals prevents the emergence of another major outbreak, as observed in the case of the k, λ control. Using this intuition, the region of strong asymmetry should be delimited by those intervention times yielding the highest epidemic peaks for both interventions, i.e., τ_μ^* and $\tau_{k,\lambda}^*$. Thus, we can compute them with Eq. (11), using the α value of each policy. The red dashed lines in Fig. 3(a) show a fair agreement between our theoretical bounds for this region and the results from numerical simulations.

Figure 3(a) also illustrates that longer interventions and moderate evolutionary rates revert the asymmetric outcome of control policies, producing a higher peak for the μ control strategy, i.e., $\Delta i_{\max} \leq 0$. To understand this phenomenon, we consider $\tau = 200$ and represent the epidemic and evolutionary curves assuming $\mathcal{D} = 10^{-8}$ [Fig. 3(d)] and $\mathcal{D} = 10^{-7}$ [Fig. 3(e)]. In both cases, $s(t)$ does not allow us to explain the unequal outcome, as the pool of susceptible individuals is only partially depleted and of similar size for each strategy. We should instead focus on how the different control policies shape the evolution of the virus. While μ control yields the same early-time evolution of $\mathcal{R}_0$ as in the uncontrolled case, k, λ control slows down viral evolution. Equation (8) captures this difference, encoded in the parameter α , with a fair agreement between the theoretical lines and the numerical results at early stages of the outbreak.

Therefore, for intermediate evolutionary speeds, i.e., $\mathcal{D} \sim 10^{-7}$, the pool of susceptible agents is large and of similar size across strategies at $t = \tau$. However, viruses under μ control have much higher infectiousness than those under k, λ control, thus explaining the higher epidemic peak in the secondary outbreak [36]. Note that this behavior does not appear for faster evolutionary rates as they induce a much greater depletion of the susceptible population, not observing any secondary peak.

Conclusions—We have shown that allowing pathogen infectivity to evolve reshapes in a fundamental way both epidemic dynamics and the effectiveness of control strategies. The interplay between contagion (selection) and mutation (diffusion) induces superexponential early-time prevalence growth. This dynamical behavior has been theoretically predicted for epidemics [31] and reported in empirical data for influenza outbreaks [37], whereby it was attributed to a different underlying mechanism. Accelerated growth rates have also been observed in other biological dynamics [38] such as tumor growth [39] or bacterial proliferation [40]. The model also displays abrupt epidemic transitions [31], positioning trait evolution as an additional pathway toward explosive transitions in contagion dynamics [41–43].

When control measures are incorporated, these evolutionary effects lead to qualitatively different and counter-intuitive outcomes. In particular, both the attack rate and the epidemic peak depend nonmonotonically on intervention duration, implying that prematurely lifted interventions may worsen epidemic outcomes relative to doing nothing. This nonmonotonic behavior is unimodal for the epidemic peak but presents multiple local maxima for the attack rate (see Supplemental Material Note 5 [35]); therefore, multiple epidemiological indicators should be computed for a complete understanding of how control policies shape the impact and evolution of epidemic outbreaks [14].

Furthermore, we identified an asymmetry between control strategies: interventions acting on transmission parameters and those shortening the infectious period are no longer equivalent in the presence of infectivity evolution. Shortening the infectious period, e.g., by administering drugs, accelerates epidemic timescales without slowing infectivity evolution, whereas acting on interhost transmission slows both prevalence and infectivity growth. As a result, the relative effectiveness of control strategies may be reversed: policies that appear optimal in non-evolving settings can become counterproductive once evolutionary effects are taken into account, particularly when interventions are lifted after the first epidemic peak.

Overall, using this minimal model, we have revealed a natural multiscale coupling between mutation dynamics within hosts and epidemic control at the population level, which determines the selection of newly emerging variants and, consequently, viral evolution. Despite the simplicity of the model, our findings are quite general and expected to appear in more biologically grounded models including complementary evolutionary pathways such as antigenic drift [32,44–46] or time-evolving recovery rates, exhaustive descriptions of within-host dynamics [47,48], or evolutionary trade-offs constraining viral evolution [49–51].

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Data availability—The code to reproduce the results reported in this Letter is available at [52].

- [1] M. J. Keeling and P. Rohani, *Modeling Infectious Diseases in Humans and Animals* (Princeton university press, Princeton, NJ, 2008).
- [2] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, and A. Vespignani, Epidemic processes in complex networks, *Rev. Mod. Phys.* **87**, 925 (2015).
- [3] W. O. Kermack and A. G. McKendrick, A contribution to the mathematical theory of epidemics, *Proc. R. Soc. A* **115**, 700 (1927).
- [4] H. W. Hethcote, The mathematics of infectious diseases, *SIAM Rev.* **42**, 599 (2000).
- [5] S. Flaxman, S. Mishra, A. Gandy, H. J. T. Unwin, T. A. Mellan, H. Coupland, C. Whittaker, H. Zhu, T. Berah, J. W. Eaton *et al.*, Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe, *Nature (London)* **584**, 257 (2020).
- [6] S. Hsiang, D. Allen, S. Annan-Phan, K. Bell, I. Bolliger, T. Chong, H. Druckenmiller, L. Y. Huang, A. Hultgren, E. Krasovich *et al.*, The effect of large-scale anti-contagion policies on the COVID-19 pandemic, *Nature (London)* **584**, 262 (2020).
- [7] S. M. Kissler, C. Tedijanto, E. Goldstein, Y. H. Grad, and M. Lipsitch, Projecting the transmission dynamics of SARS-CoV-2 through the postpandemic period, *Science* **368**, 860 (2020).
- [8] A. Arenas, W. Cota, J. Gómez-Gardeñes, S. Gómez, C. Granell, J. T. Matamalas, D. Soriano-Paños, and B. Steinegger, Modeling the spatiotemporal epidemic spreading of COVID-19 and the impact of mobility and social distancing interventions, *Phys. Rev. X* **10**, 041055 (2020).
- [9] F. Di Lauro, I. Z. Kiss, and J. C. Miller, Optimal timing of one-shot interventions for epidemic control, *PLoS Comput. Biol.* **17**, e1008763 (2021).
- [10] P.-A. Bliman, M. Duprez, Y. Privat, and N. Vauchelet, Optimal immunity control and final size minimization by social distancing for the SIR epidemic model, *J. Optim. Theory Appl.* **189**, 408 (2021).
- [11] D. I. Ketcheson, Optimal control of an SIR epidemic through finite-time non-pharmaceutical intervention, *J. Math. Biol.* **83**, 7 (2021).
- [12] T. A. Perkins and G. España, Optimal control of the COVID-19 pandemic with non-pharmaceutical interventions, *Bull. Math. Biol.* **82**, 118 (2020).
- [13] J. Hindes and I. B. Schwartz, Epidemic extinction and control in heterogeneous networks, *Phys. Rev. Lett.* **117**, 028302 (2016).
- [14] E. Rozán, M. N. Kuperman, and S. Bouzat, Analysis of non pharmaceutical interventions with SIR epidemic models: Decreasing the infection peak vs. minimizing the epidemic size, *arXiv:2604.08420*.
- [15] D. H. Morris, F. W. Rossine, J. B. Plotkin, and S. A. Levin, Optimal, near-optimal, and robust epidemic control, *Commun. Phys.* **4**, 78 (2021).
- [16] B. T. Grenfell, O. G. Pybus, J. R. Gog, J. L. Wood, J. M. Daly, J. A. Mumford, and E. C. Holmes, Unifying the epidemiological and evolutionary dynamics of pathogens, *Science* **303**, 327 (2004).
- [17] R. M. Anderson and R. M. May, *Infectious Diseases of Humans: Dynamics and Control* (Oxford University Press, Oxford, 1991).
- [18] T. Day and S. Gandon, Applying population-genetic models in theoretical evolutionary epidemiology, *Ecol. Lett.* **10**, 876 (2007).
- [19] S. Lion and J. A. Metz, Beyond R_0 maximisation: On pathogen evolution and environmental dimensions, *Trends Ecol. Evol.* **33**, 458 (2018).
- [20] L. S. Tsimring, H. Levine, and D. A. Kessler, RNA virus evolution via a fitness-space model, *Phys. Rev. Lett.* **76**, 4440 (1996).
- [21] Y. Iwasa, F. Michor, and M. A. Nowak, Evolutionary dynamics of escape from biomedical intervention, *Proc. R. Soc. B* **270**, 2573 (2003).
- [22] Y. Iwasa, F. Michor, and M. A. Nowak, Evolutionary dynamics of invasion and escape, *J. Theor. Biol.* **226**, 205 (2004).
- [23] M. Hartfield and S. Alizon, Within-host stochastic emergence dynamics of immune-escape mutants, *PLoS Comput. Biol.* **11**, e1004149 (2015).
- [24] D. Van Egeren, A. Novokhodko, M. Stoddard, U. Tran, B. Zetter, M. S. Rogers, D. Joseph-McCarthy, and A. Chakravarty, Controlling long-term SARS-CoV-2 infections can slow viral evolution and reduce the risk of treatment failure, *Sci. Rep.* **11**, 22630 (2021).
- [25] M. Hartfield and S. Alizon, Epidemiological feedbacks affect evolutionary emergence of pathogens, *Am. Nat.* **183**, E105 (2014).
- [26] M. T. Meehan, R. C. Cope, and E. S. McBryde, On the probability of strain invasion in endemic settings: Accounting for individual heterogeneity and control in multi-strain dynamics, *J. Theor. Biol.* **487**, 110109 (2020).
- [27] B. Ashby, C. A. Smith, and R. N. Thompson, Non-pharmaceutical interventions and the emergence of pathogen variants, *Evol. Med. Public Health* **11**, 80 (2023).
- [28] S. Dellicour, G. Baele, G. Dudas, N. R. Faria, O. G. Pybus, M. A. Suchard, A. Rambaut, and P. Lemey, Phylodynamic assessment of intervention strategies for the West African Ebola virus outbreak, *Nat. Commun.* **9**, 2222 (2018).
- [29] Z. Chen, J. L.-H. Tsui, B. Gutierrez, S. Busch Moreno, L. Du Plessis, X. Deng, J. Cai, S. Bajaj, M. A. Suchard, O. G. Pybus *et al.*, COVID-19 pandemic interventions reshaped the global dispersal of seasonal influenza viruses, *Science* **386**, eadq3003 (2024).
- [30] Z. Chen, J. L.-H. Tsui, J. Cai, S. Su, C. Viboud, L. Du Plessis, P. Lemey, M. U. Kraemer, and H. Yu, Disruption of seasonal influenza circulation and evolution during the 2009 H1N1 and COVID-19 pandemics in Southeastern Asia, *Nat. Commun.* **16**, 475 (2025).
- [31] X. Zhang, Z. Ruan, M. Zheng, J. Zhou, S. Boccaletti, and B. Barzel, Epidemic spreading under mutually independent intra- and inter-host pathogen evolution, *Nat. Commun.* **13**, 6218 (2022).
- [32] D. Soriano-Paños, Eco-evolutionary constraints for the endemicity of rapidly evolving viruses, *PRX Life* **3**, 043001 (2025).

- [33] I. M. Rouzine and G. Rozhnova, Antigenic evolution of viruses in host populations, *PLoS Pathogens* **14**, e1007291 (2018).
- [34] V. Chardès, A. Mazzolini, T. Mora, and A. M. Walczak, Evolutionary stability of antigenically escaping viruses, *Proc. Natl. Acad. Sci. U.S.A.* **120**, e2307712120 (2023).
- [35] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/56yd-sfks> for additional details of the model, the robustness analysis of the nonmonotonic impact of control policies for different models of evolution, and further analytical insights into the origin of this phenomenon.
- [36] P. Castioni, S. Gómez, C. Granell, and A. Arenas, Rebound in epidemic control: How misaligned vaccination timing amplifies infection peaks, *npj Complex* **1**, 20 (2024).
- [37] S. V. Scarpino, A. Allard, and L. Hébert-Dufresne, The effect of a prudent adaptive behaviour on disease transmission, *Nat. Phys.* **12**, 1042 (2016).
- [38] R. Durrett, J. Foo, K. Leder, J. Mayberry, and F. Michor, Evolutionary dynamics of tumor progression with random fitness values, *Theor. Popul. Biol.* **78**, 54 (2010).
- [39] B. Ocaña-Tienda, J. Pérez-Beteta, J. Jiménez-Sánchez, D. Molina-García, A. Ortiz de Mendivil, B. Asenjo, D. Albillo, L. A. Pérez-Romasanta, M. Valiente, L. Zhu *et al.*, Growth exponents reflect evolutionary processes and treatment response in brain metastases, *npj Syst. Biol. Appl.* **9**, 35 (2023).
- [40] A. Cylke and S. Banerjee, Super-exponential growth and stochastic size dynamics in rod-like bacteria, *Biophys. J.* **122**, 1254 (2023).
- [41] R. M. D'Souza, J. Gómez-Gardenes, J. Nagler, and A. Arenas, Explosive phenomena in complex networks, *Adv. Phys.* **68**, 123 (2019).
- [42] S. Lamata-Otín, J. Gómez-Gardeñes, and D. Soriano-Paños, Pathways to discontinuous transitions in interacting contagion dynamics, *J. Phys. Complex.* **5**, 015015 (2024).
- [43] S. Lamata-Otín, A. Reyna-Lara, D. Soriano-Paños, V. Latora, and J. Gómez-Gardeñes, Collapse transition in epidemic spreading subject to detection with limited resources, *Phys. Rev. E* **108**, 024305 (2023).
- [44] A. Sasaki, S. Lion, and M. Boots, Antigenic escape selects for the evolution of higher pathogen transmission and virulence, *Nat. Ecol. Evol.* **6**, 51 (2022).
- [45] S. Lamata-Otín, O. C. Rotita-Ion, A. Arenas, D. Soriano-Paños, and J. Gómez-Gardeñes, Genotype networks drive oscillating endemicity and epidemic trajectories in viral evolution, *Commun. Phys.* **8**, 502 (2025).
- [46] B. J. Williams, G. St-Onge, and L. Hébert-Dufresne, Localization, epidemic transitions, and unpredictability of multi-strain epidemics with an underlying genotype network, *PLoS Comput. Biol.* **17**, e1008606 (2021).
- [47] N. Mideo, S. Alizon, and T. Day, Linking within- and between-host dynamics in the evolutionary epidemiology of infectious diseases, *Trends Ecol. Evol.* **23**, 511 (2008).
- [48] F. Fabre, J. Montarry, J. Coville, R. Senoussi, V. Simon, and B. Moury, Modelling the evolutionary dynamics of viruses within their hosts: A case study using high-throughput sequencing, *PLoS Pathogens* **8**, e1002654 (2012).
- [49] S. Alizon, Transmission-recovery trade-offs to study parasite evolution, *Am. Nat.* **172**, E113 (2008).
- [50] J. C. De Roode, A. J. Yates, and S. Altizer, Virulence-transmission trade-offs and population divergence in virulence in a naturally occurring butterfly parasite, *Proc. Natl. Acad. Sci. U.S.A.* **105**, 7489 (2008).
- [51] M. A. Acevedo, F. P. Dilleuth, A. J. Flick, M. J. Faldyn, and B. D. Elder, Virulence-driven trade-offs in disease transmission: A meta-analysis, *Evolution* **73**, 636 (2019).
- [52] S. Lamata-Otín, SIRevolution, <https://github.com/santiago-laot/SIRevolution> (2026).

End Matter

Appendix A: Sigmoid approximation for susceptible depletion—To analyze the slowdown in the growth of the average infectivity induced by susceptible depletion, we approximate the susceptible fraction by a sigmoid function centered at the epidemic peak, $s(t) \simeq s(\infty) + (1 - s(\infty))/(1 + e^{\sigma(t-t_p)})$. Substituting this expression into Eq. (5) and integrating yields an explicit approximation for the evolution of the average infectivity at intermediate times (see Supplemental Material Note 7 [35]). Retaining the dominant contribution for times around and beyond the peak time t_p , we obtain

$$\bar{\lambda}(t) \sim kD \left[s(\infty) \frac{t^2}{2} - (1 - s(\infty)) \left(\frac{\sigma t + 1}{\sigma^2} e^{-\sigma(t-t_p)} \right) \right], \quad (\text{A1})$$

which explicitly displays a negative transient contribution that temporarily hinders the growth of infectivity [see Fig. 1(c)]. As time increases, the exponential term vanishes and the dynamics smoothly crosses over to a

quadratic growth controlled by the remaining susceptible fraction $s(\infty)$.

Appendix B: Hybrid approximation to derive the epidemic peak—To assess the impact of infectivity evolution on the effectiveness of control policies, we adopt a hybrid approximation in which the epidemic dynamics during the intervention window $t \in [0, \tau]$ follows the early-time controlled evolution of $i(t, \varepsilon)$ and $\bar{\lambda}(t, \varepsilon)$, while after lifting the intervention the outbreak evolves as an uncontrolled SIR process, with initial conditions given by the state at $t = \tau$ (see Supplemental Material Note 4 [35] for details).

Under this approximation, and assuming that the epidemic peak occurs after lifting the intervention, the post-intervention peak can be approximated using the standard SIR result [43]

$$i_{\max}(\tau) \approx 1 - r(\tau) - \frac{1}{\mathcal{R}_0(\tau)} [1 + \ln(\mathcal{R}_0(\tau)s(\tau))], \quad (\text{B1})$$

where $\mathcal{R}_0(\tau) = k\bar{\lambda}(\tau)/\mu$ and $s(\tau) = 1 - i(\tau) - r(\tau)$. Note that this expression is a strong assumption as it neglects the impact of evolution on the epidemic peak; evolution just affects the initial conditions when lifting the intervention.

Differentiating Eq. (B1) with respect to τ and using the expressions for the controlled dynamics at the lifting time [Eqs. (8) and (9)] yields the general identity reported in Eq. (10), which explicitly separates the direct epidemiological contribution from the evolutionary one.

Appendix C: Critical effective diffusion strength—To quantify when infectivity evolution reverses the monotonic dependence of the epidemic peak on the intervention duration, we analyze the condition $di_{\max}/d\tau = 0$, reported in Eq. (10) and derived in Appendix B. Setting $di_{\max}/d\tau = 0$ defines an implicit relation for the critical effective diffusion strength,

$$\mathcal{D}_c(\tau) = \frac{\mu^2(1-\varepsilon)i(\tau)\mathcal{R}_0(\tau)^2}{\varepsilon k^2 \tau \ln(\mathcal{R}_0(\tau)s(\tau))}. \quad (\text{C1})$$

Assuming that the epidemic prevalence remains small at the lifting time, $i(\tau) \ll 1$, we approximate $s(\tau) \simeq 1$ and $\ln(\mathcal{R}_0(\tau)s(\tau)) \simeq \ln \mathcal{R}_0$, and evaluate the logarithmic and

prefactor terms at their zeroth-order values, $\mathcal{R}_0(\tau) \simeq \mathcal{R}_0$, while retaining only the leading dependence on $\mathcal{D}$ through the exponential contribution in $i(\tau)$. As detailed in Supplemental Material Note 4 [35], under these approximations and using Eq. (9), Eq. (C1) reduces to a transcendental equation that admits a closed-form solution in terms of the Lambert W function,

$$\mathcal{D}_c(\tau) = -\frac{1}{P(\tau)} W_0[-P(\tau)Q(\tau)], \quad (\text{C2})$$

where $P(\tau) = \alpha^2 k^2 \tau^3 / 6$ and $Q(\tau) = (\mu^2(1-\varepsilon)\mathcal{R}_0^2) / (\varepsilon k^2 \tau \ln \mathcal{R}_0) i_0 e[\mu\alpha(\varepsilon\mathcal{R}_0 - 1)\tau/\varepsilon]$. In the expression above $\alpha = \varepsilon$ for k or λ control and $\alpha = 1$ for μ control.

Equation (C2) exhibits a divergence $\mathcal{D}_c(\tau) \rightarrow \infty$ as $\tau \rightarrow 0$, reflecting the fact that the hybrid approximation accounts for infectivity evolution only during the controlled phase and therefore does not capture the $\tau \rightarrow 0$ limit. For finite intervention durations, however, it predicts a critical effective diffusion strength such that, for $\mathcal{D} > \mathcal{D}_c(\tau)$ the epidemic peak increases with the intervention duration, whereas for $\mathcal{D} < \mathcal{D}_c(\tau)$ it decreases.